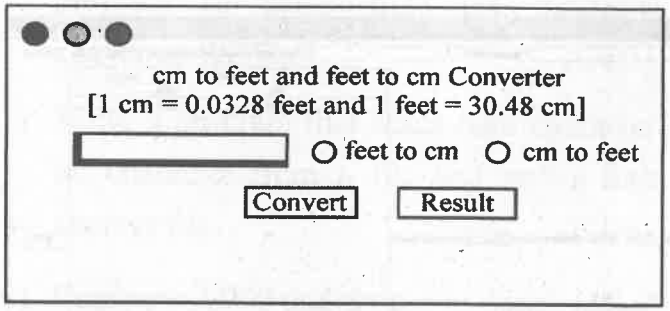
sequence only. The final result will be replaced at Result label : (CO5)



(c) Write a Java program in which data of employees has to be inserted in Employee table in EmployeeDatabase database. Also explain the use of JDBC-MySQL connection in this program. Make sure to display the details of the employee inserted into the database : (CO5)

Table	Employye (id int, name varchar (255), department varchar (255), salary varchar (255))
Database	Employee Database
User	Root
Password	Root
Connector	jdbc;mysql

H **Roll No. .....**

TCS–408

B. TECH. (CSE)

(FOURTH SEMESTER)

END SEMESTER EXAMINATION,

June, 2023

JAVA PROGRAMMING LANGUAGE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the importance of String literals with the help of examples. Also, compare StringBuffer and StringBuilder. (CO1)

(b) The Employee has the following attributes :

ID, Name, Department, City and Salary

Write a program that uses concept of constructor chaining to construct the objects. Also discuss importance of using constructor chaining in a class. (CO1)

(c) International Standard Book Number (ISBN) is a unique numeric book identifier which is printed on every book. The ISBN is based up on 10-digit code.

The ISBN is legal, if :

$1 \times digit_1 + 2 \times digit_2 + 3 \times digit_3 + 4 \times digit_4 + 5 \times digit_5 + 6 \times digit_6 + 7 \times digit_7 + 8 \times digit_8 + 9 \times digit_9 + 10 \times digit_{10}$ is divisible by 11.

Example :

Input : 1401601499

Sum = $1 \times 1 + 2 \times 4 + 3 \times 0 \times 4 \times 1 + 5 \times 6 + 6 \times 0 + 7 \times 1 + 8 \times 4 + 9 \times 9 + 10 \times 9 = 253$

which is divisible by 11.

Write a Java program to check whether the entered ISBN number is valid or not using following conditions :

(i) If the ISBN is not a 10-digit integer, output the message, "Illegal ISBN' and terminate the program.

(ii) If the number is 10-digit, extract the digits of the number and compute the sum as explained.

If the sum is divisible by 11, output the message, "Legal ISBN" otherwise "Illegal ISBN".

2. (a) A programmer wants to keep all similar classes and interfaces in same package. So programmer has created a package "package1" having one class "C1", one Interface "I1" and one subpackage "package2". The "package2" contains class "C2". The details of members are given below :

Class "C1" contains mt sum (int, int) method to calculate sum of two numbers.

Interface I1 contains an abstract method mt sub (int, int) to subtract two numbers.

The programmer wants to reuse "C1" and "I1" in class "C2" which is placed inside subpackage "package2" for sum and subtraction. Write a program to implement the given scenario. (CO2)

(b) Why is multiple inheritance using classes not allowed in Java ? Discuss with an example. Also discuss various ways of abstraction with codes. (CO2)

(c) Discuss the importance of the super keyword in Java. A superclass "Worker" has been defined to store the details of a worker, having name and basic wage of the worker. A constructor is used to initialize the values of the class, and the method display() is used to display the name and basic wage.

Another subclass, "Wages" is defined to compute the monthly wages for the worker. If a worker works for 'h' hours overtime, then he/she is paid 'r' rupees per

hour. Calculate the monthly wage he receives at the month's end (assuming a month has 30 days). (CO3)

3. (a) A bank's client has the following attributes and methods : (CO3)

Instance variables : name, account_no, amount

Instance methods : void withdraw(int) to withdraw some amount from the account. A user wants to withdraw some money from his account. Create a LowBalanceException that occurs when the user tries to withdraw an amount that is greater than his current bank balance. This exception will print "Your account has a low balance ....."

Example: name="xyz"

account_no=101

amount=1 000

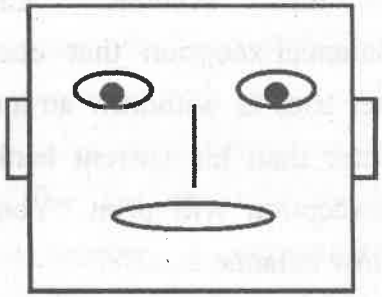
If client will try to withdraw 5000 INR, then output is :

"Your account has a low balance....."

(b) Discuss the life cycle of a thread with a neat and clean diagram. Also, write a program to demonstrate the producer consumer problem. (CO3)

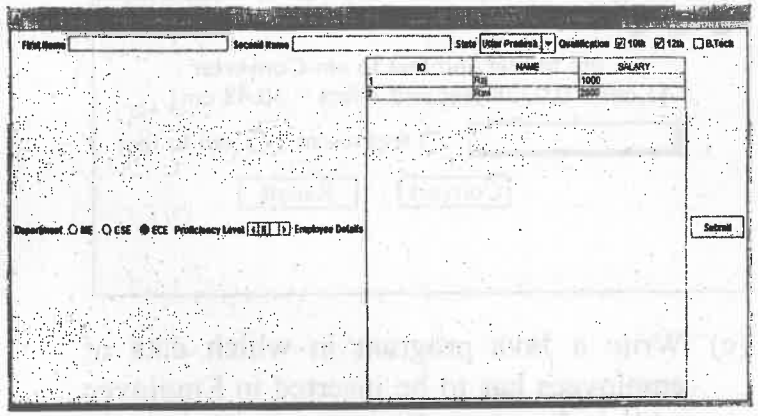
(c) Write a program that reads data character by character from a file and writes into another file. (CO3)

4. (a) Compare AWT and Swing in detail. Also, write a swing program to create the following share : (CO4)



(b) Discuss the importance of Collection and Generic Framework in Java. Write a Java code to insert 90, 15 at the end and 45 at index 2 of an ArrayList<Integer> having elements 10, 22, 43, 64, 95. (CO4)

(c) Write a Java program to design the following output : (CO4)



5. (a) Discuss JDBC drivers with neat and clean diagrams. Also discuss the various steps and corresponding lines of code when you want to use JDBC in your program. (CO5)

(b) Create the given application using AWT/Swing, in which you create a converter which has two functionalities [feet to cm and cm to feet]. Frame must include information about the conversion as shown in fig. below. Order of the buttons and fields must be in given